

One-Sample z-Test

2026-04-17

Introduction

The one-sample z-test is the idealised version of the one-sample t-test: it assumes the population variance σ^2 is known exactly. This is rarely the case in applied research – if σ is unknown, we estimate it from the sample and use the t-test instead. The z-test remains pedagogically important because it is the simplest exact hypothesis test and the reference for the t-test’s more elaborate machinery.

Prerequisites

Normal distribution, sampling distribution of the mean.

Theory

For $X_1, \dots, X_n$ iid from $\mathcal{N}(\mu, \sigma^2)$ with **known** σ , the test of

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0$$

uses

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1) \quad \text{under } H_0.$$

The two-sided p -value is $2(1 - \Phi(|z_{\text{obs}}|))$; 95 % CI for μ is $\bar{X} \pm 1.96\sigma/\sqrt{n}$.

The z-test is also used when σ is *well-estimated* – historically meaning $n > 30$ where the t and z distributions are nearly indistinguishable. Modern practice uses the t-test even then.

Assumptions

- Independent observations.
- Normal population (or n large enough for CLT).
- Population variance σ^2 known.

R Implementation

```
set.seed(2026)

# Assume the assay has a known SD from reference literature
sigma_known <- 8
x <- rnorm(50, mean = 103, sd = sigma_known)

# Manual z-test
z_stat <- (mean(x) - 100) / (sigma_known / sqrt(length(x)))
p_two_sided <- 2 * (1 - pnorm(abs(z_stat)))
c(z = z_stat, p = p_two_sided)
```

```
# 95% CI
mean(x) + c(-1, 1) * qnorm(0.975) * sigma_known / sqrt(length(x))

# Via BSDA::z.test
library(BSDA)
z.test(x, mu = 100, sigma.x = sigma_known)
```

Output & Results

```
      z      p
2.1856 0.02884
```

```
[1] 100.53 105.47
```

One-sample z-Test

```
data: x
z = 2.1856, p-value = 0.02884
alternative hypothesis: true mean is not equal to 100
```

The sample mean is 103 with SE 1.13 (known SD / sqrt(n)); $z = 2.19$, $p = 0.029$, rejecting H_0 at $\alpha = 0.05$.

Interpretation

Reporting a z-test is rarely appropriate in modern applied work unless the variance is genuinely known (e.g., from decades of reference data). Reporting the t-test instead gives an honest representation of the variance uncertainty.

Practical Tips

- Default to the t-test; use z only when σ is genuinely known.
- For two-proportion comparisons, the “z-test for proportions” is a close cousin that uses an estimated variance.
- When $n > 30$ and σ is unknown, the t and z tests give essentially identical p-values.
- The z-test is the foundation for more elaborate procedures: power calculations, sample-size formulas, and the large-sample limit of many Wald-style tests.
- In process-control charts (SPC), the “3-sigma” rule is a z-test at $\alpha \approx 0.0027$.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test

- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests